

Python

A quickstart into the key concepts of programming

Basic syntax

Key concepts in programming

- Variables (*integers, strings, dates, etc.*)
- Flow control (*if then, loop, etc.*)
- Functions (*named list of steps the code will follow, with clear input/output*)

Python: basic syntax

basic_syntax.ipynb

Python files

- Python scripts are saved with `.py`
- Jupyter notebook files: `.ipynb`
- Naming convention
 - lowercase, with words separated by underscores as necessary to improve readability
 - Indentation: 4 spaces (not tabs!)
 - Max line length: 79 characters
 - <https://peps.python.org/pep-0008/>
 - https://visualgit.readthedocs.io/en/latest/pages/naming_convention.html

Python syntax

- *Syntax:*
 - refers to the structure of the language
 - i.e., what constitutes a correctly formed program.
 - E.g, 'country' is correct, but 'Kountri' is syntactically wrong
- *Semantics:*
 - involve the meaning of the statements
 - E.g., 'humans swim' is semantically correct, but 'humans fly' is semantically wrong (although syntactically correct!)

Your Python code should be syntactically AND semantically correct (for correct scientific results and reproducibility).

Comment

- Comments are marked by `#`
- Anything on the line following the hash sign is ignored by the interpreter
- `x += 2 # shorthand for x = x + 2`
- multiline comments:
 - no syntax out of the box
 - use consecutive `#` single-line comments
 - Python-aware editor can help

Statement

- A statement is one complete *sentence* in the language. It contains one complete instruction.
- End-of-Line terminates a statement (no semicolon required)

```
B = A
```

```
surfC = math.pi * r**2
```

- Continuation: continue to the next line, use the \ marker

```
x = 1 + 2 + 3 + 4\  
    + 5 + 6 + 7 + 8
```

- Semicolon can optionally terminate a statement (discouraged)

```
lower = []; upper = []
```

Code blocks

- newlines and indentation are syntax in python.
- Statements can be grouped into **blocks**
 - a *block* of code is a set of statements that should be treated as a unit.
 - Structures that introduce a block, end with a colon :
 - Many operations define a **code block**
 - Code to run within a loop
 - Code to run (or not) depending on a logical test

Code blocks

- Blocks are indicated by **indentation level**.
 - Indentation is mandatory in Python and must be exact.
 - Python style recommends **4 spaces** for indentation
 - Indent each block by however many spaces (at least 1) you wish to use, but each block level must be indented by exactly the same number.
 - Do not use tabs, spaces are the preferred indentation method.
 - Python-aware editors (e.g. Spyder, VSCode, vim) will automatically indent the next statement to the same level as the one before it; so, check your first indent
 - Check your IDE for defaults

Indentation: Whitespace matters

- In C, code blocks are denoted by curly braces:

```
// C code
for(int i=0; i<100; i++)
{
    total += i;
}
```

- In Python, code blocks are denoted by *indentation*:

```
for i in range(100):
    # indentation indicates code block
    total += i
```

- In Python, indented code blocks are always preceded by a colon (:) on the previous line.

Indentation

```
>>> if x < 4:  
...     y = x * 2  
...     print(x)
```

- `print(x)` is in the indented block, and will be executed only if `x` is less than 4.
- *File: check_whitespace.py*

```
>>> if x < 4:  
...     y = x * 2  
... print(x)
```

- `print(x)` is outside the block, and will be executed regardless of the value of `x`,



Code blocks

- There must be at least 1 statement in a block.
 - You can use `pass` as a way to make a dummy statement if you really need to.
 - When you do not know yet what code to write, put a `pass` at that line.

```
n = 10
# use pass inside if statement
if n > 10:
    pass
print('Hello')
```

- File: `pass_example.py`

Whitespace

- *amount* of whitespace used for indenting code blocks is up to the user, as long as it is consistent throughout the script.
- Whitespace *within* lines does not matter

```
In [4]: x=1+2
```

```
x = 1 + 2    # recommended by PEP-8
```

```
x      =      1      +      2
```